Robot Learning

Lecture 0. A few info

What is (robot) learning

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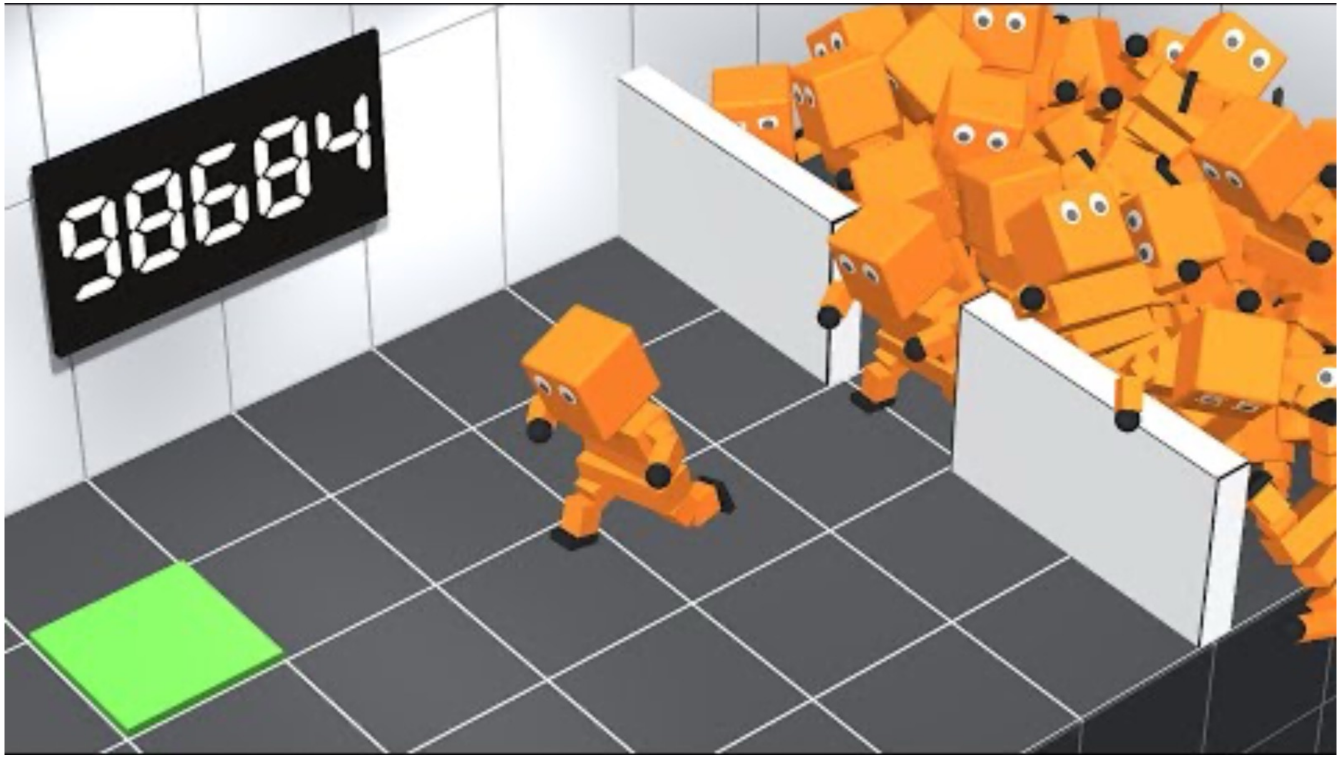

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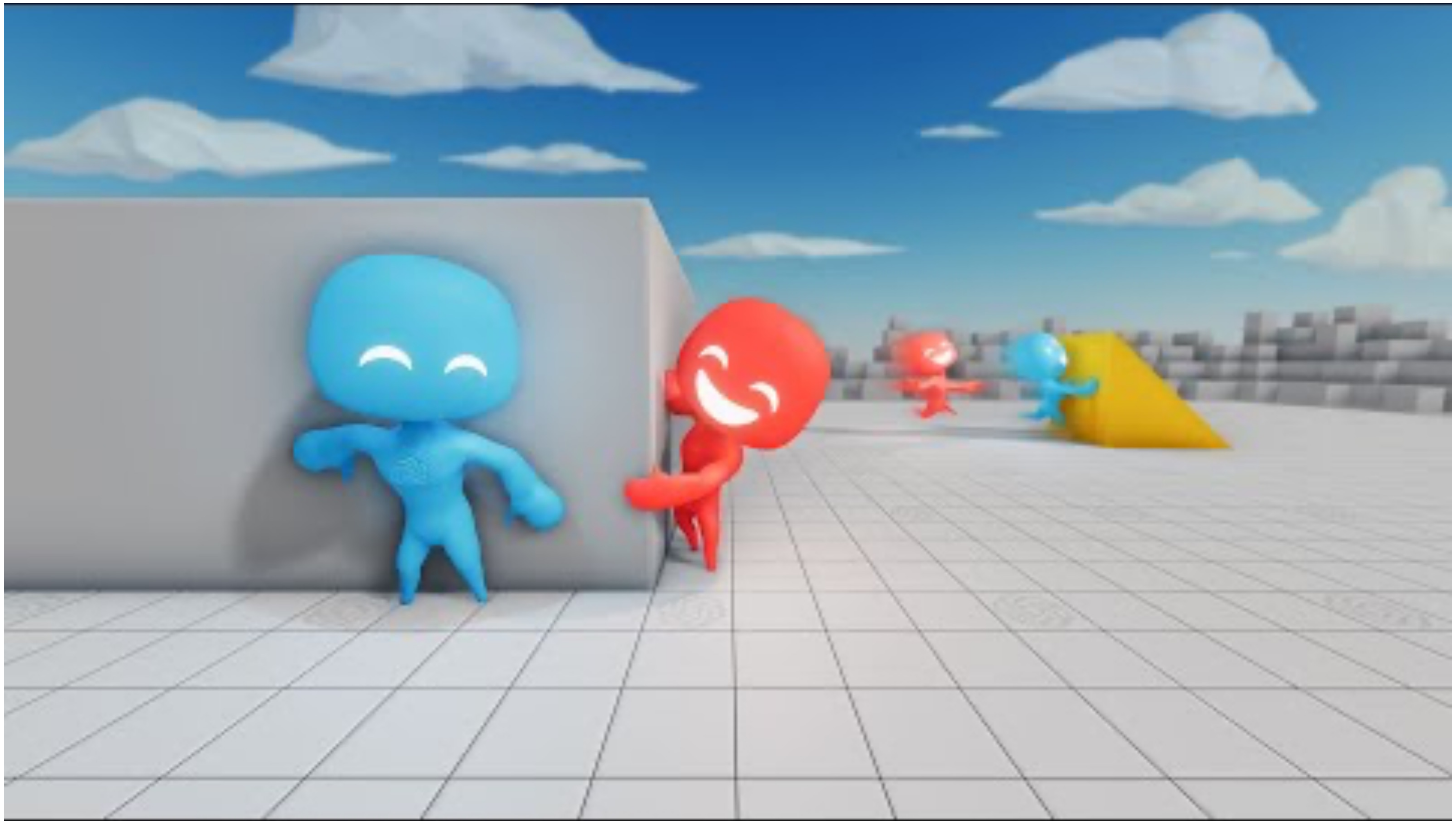

Teachers

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Approx. 30 h

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PhD student @ PoliTo (III y)

Davide Buoso

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Approx. 15 h

PhD student @ PoliTo (I y)

Topics

- Mainly reinforcement learning → learning from experience
- Often applied to robotic tasks
  - but the theory is general, and we will see applications in other domains
- A quick introduction to robotic systems, to provide useful tools
- Some pills on other learning paradigms for decision making (CV, scene graphs)

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- Knowledge of the main components of robotics applications, from sensing, control and actuation, to high level action decision and planning;
- Knowledge on how to formalize an autonomous learning problem, from the mathematical foundations of the optimization objectives, to the implementation of specific use-cases;
- Knowledge of the main characteristics of modern reinforcement/imitation learning techniques applied to autonomous robots;

How?

- Frontal lectures (approx. 40 h)

- In-class (and at-home) labs for hands-on learning (approx. 15h in class + N at home)

What will you learn

- Knowledge of the main components of robotics applications, from sensing, control and actuation, to high level action decision and planning;
- Knowledge on how to formalize an autonomous learning problem, from the mathematical foundations of the optimization objectives, to the implementation of specific usecases;
- Knowledge of the main characteristics of modern reinforcement learning techniques applied to autonomous robots;
- Basic knowledge on how to deal with research questions, how to read a paper, how to organize a project and prepare a scientific conference report.

How?

- Frontal lectures (approx. 40 h)
- In-class (and at-home) labs for hands-on learning (approx. 15h in class + N at home)
- Journal clubs → we take some time to discuss together foundational papers related to the topic of the week
- Team project to propose a personal extension of labs

Weekly schedule (at the moment)

- Monday 16-19
- Thursday 10-11.30
- Labs (when?)

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- Monday 16-19 --> Theory
- Thursday 10-11.30 --> Theory/journal club
- Labs (when?)

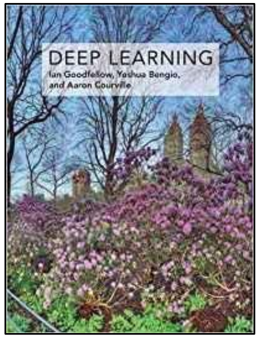
Exam - Masters' students

- Labs – 6 points
  - 4 labs MANDATORY – each 1.5 points
  - Each lab does have its own deadline – penalty for late submission: -0.5 points
- Project extension – 4 points
  - Extend the works of the labs with your personal contribution (e.g. a new method, a new reward function, changing model, … open your mind)
  - Write a report on the project (6-8 pages double column, template given)
- Oral exam – 20 points
  - Questions on the topics of the course + discussion project extension
- Active participation to journal clubs – extra 3 points

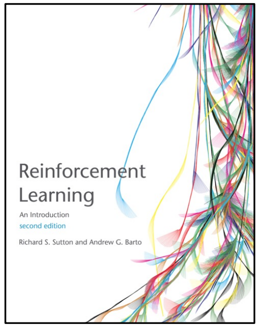
Exam - PhD students

- PhD students
  - You will work on your labs (without submission)
  - You will extend the works of the labs with your personal contribution (e.g. a new method, a new reward function, changing model, … open your mind)
  - You will submit a report on your work
  - Active participation to journal clubs, e.g. taking responsibility to introduce the paper at least once
  - Contact us and agree on alternative ways (e.g. using RL in your Ph.D. project, ...)

Some references (more on the course page)



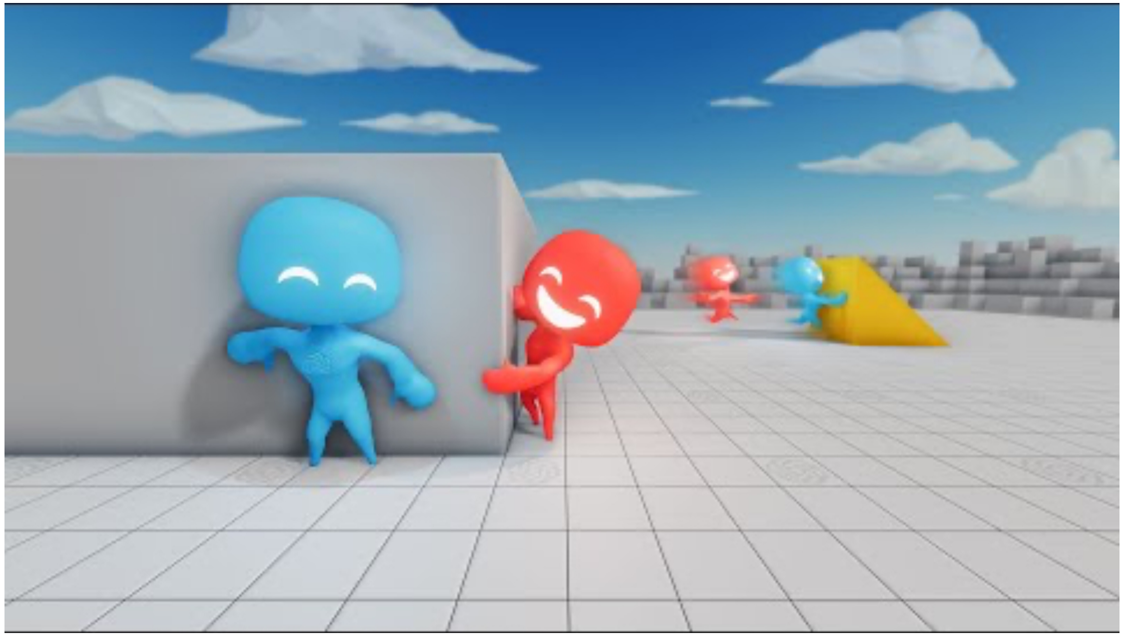
- Ian Goodfellow and Yoshua Bengio and Aaron Courville , Deep Learning, MIT Press
- Reference book for deep learning
- Freely available online



- Richard S. Sutton and Andrew G. Barto Reinforcement Learning: An Introduction, Second Edition, MIT Press
- Reference book for Reinforcement Learning
- Freely available online

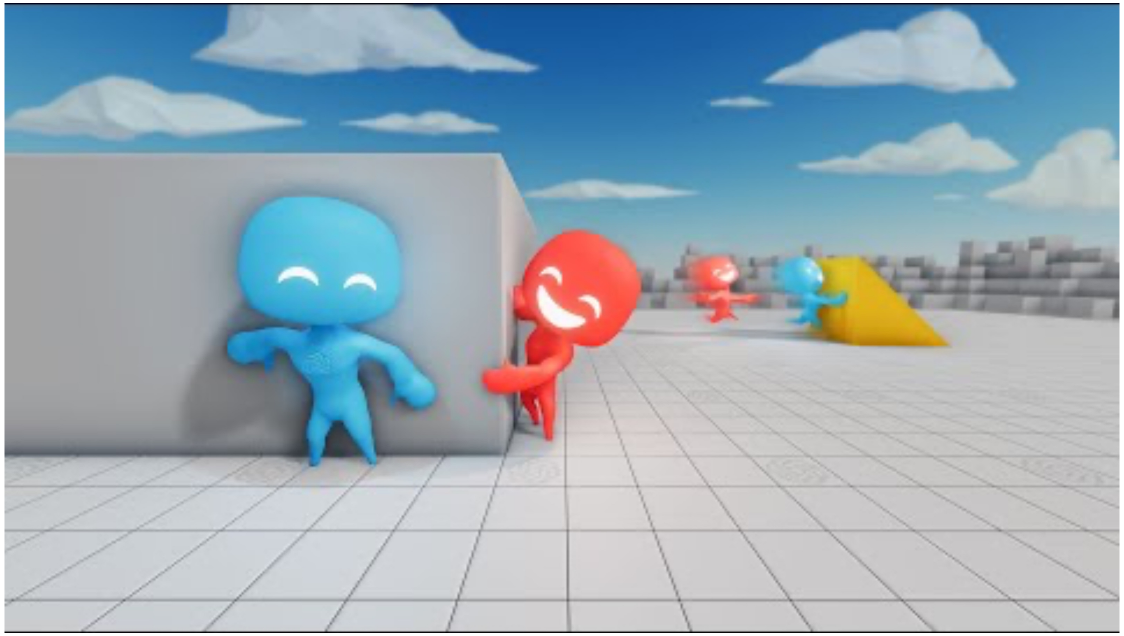
Join Telegram channel

- https://t.me/+mNPkbLmu3M85NDM0



Course website

- https://josephav.github.io/robot_learning_course/



Some references (more on the course page)

- These slides
- Papers, papers, papers!
- References in the slides
- Sutton & Barto
- Handbook of Robotics